

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture
(6th SEMESTER, 3rd Year) LAB EXPERIMENT # 01

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Score: _____ Signature of the Lab Tutor: _____ Date: _____

Basic IP Network

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Objectives

- VRRP basics and build a basic IP network as well as configure it using eNSP **Learning**

Objectives:

- Set up and navigate the eNSP simulator application.
- Establish a simple peer-to-peer network in eNSP.
- Perform capture of IP packets using Wireshark within eNSP.

Keywords: IP Network

Duration: 03 hours

1 Overview of lab

1.1. Introduction

The fundamental network behavior can be understood through the application of packet capture tools to the network. The use of Huawei's simulator platform eNSP can support

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both the implementation of technologies and the capture of packets within the network to provide a comprehensive knowledge of IP networks.

1.2. Wireshark:

A network protocol analyzer tool used to capture, inspect, and analyze network traffic in real time. It helps network engineers, cybersecurity professionals, and developers understand what's happening on a network at a very detailed level.

- Captures live network traffic or opens saved capture files
- Decodes protocols (like TCP, UDP, HTTP, DNS, etc.)

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- Filters specific packets using powerful filter expressions
- Provides detailed packet-level information
- Helps troubleshoot network issues and detect malicious activities

1.3 IP Address:

Internet Protocol address is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.

An IP address is like the home address of your device on the internet or local network. It tells other devices where to send data.

Example:

192.168.0.1

Public IP: Used on the internet, globally unique

Private IP: Used inside local networks (LAN)

Static IP: Manually assigned, does not change

Dynamic IP: Assigned by DHCP, changes over time

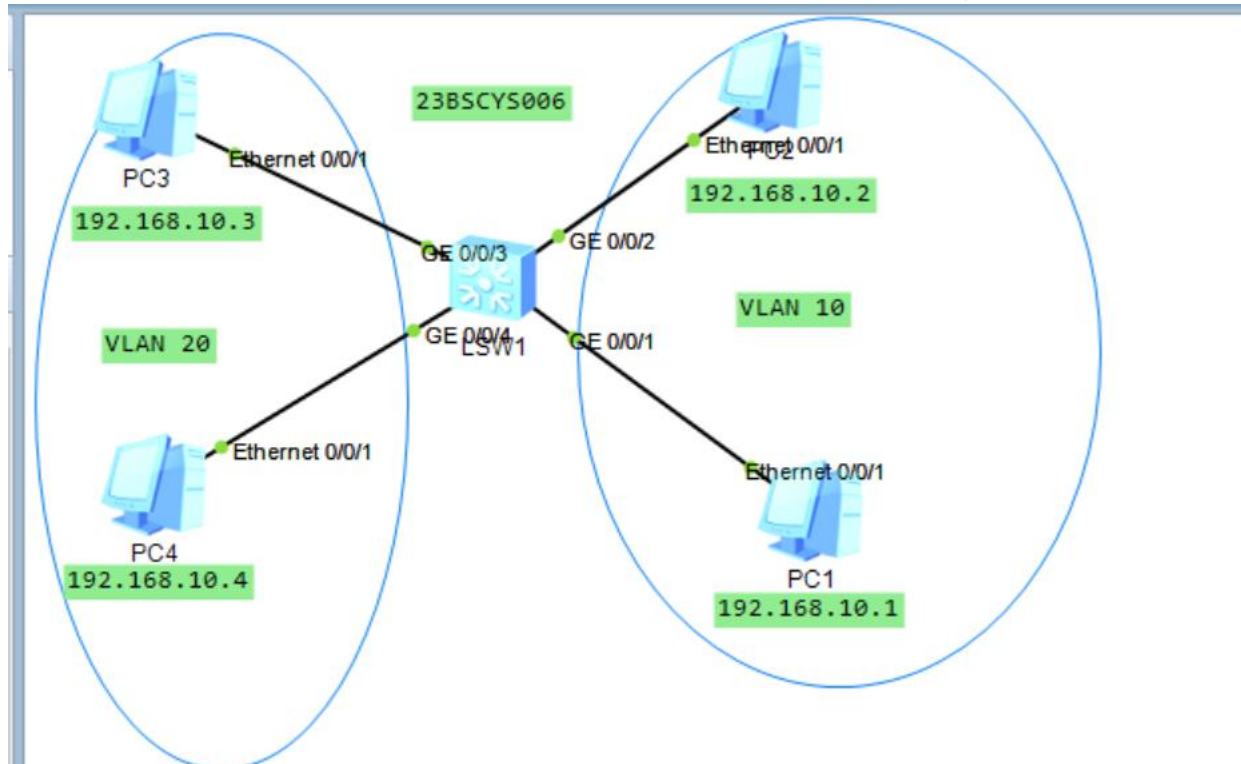
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Tasks

Try a different topology and demonstrate your learning. Try different commands.

Upload the `.topo` `.efz` `.cfg` files

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LSW1

```

(e): ETHOAM down
(dl): DLDP down
(d): Dampening Suppressed
InUti/OutUti: input utility/output utility
Interface          PHY    Protocol InUti OutUti  inErrors  outErrors
GigabitEthernet0/0/1  up    up       0%    0%      0          0
GigabitEthernet0/0/2  up    up       0%    0%      0          0
GigabitEthernet0/0/3  up    up       0%    0%      0          0
GigabitEthernet0/0/4  up    up       0%    0%      0          0
GigabitEthernet0/0/5  down  down     0%    0%      0          0
GigabitEthernet0/0/6  down  down     0%    0%      0          0
GigabitEthernet0/0/7  down  down     0%    0%      0          0
GigabitEthernet0/0/8  down  down     0%    0%      0          0
GigabitEthernet0/0/9  down  down     0%    0%      0          0
GigabitEthernet0/0/10 down  down     0%    0%      0          0
GigabitEthernet0/0/11 down  down     0%    0%      0          0
GigabitEthernet0/0/12 down  down     0%    0%      0          0
GigabitEthernet0/0/13 down  down     0%    0%      0          0
GigabitEthernet0/0/14 down  down     0%    0%      0          0
GigabitEthernet0/0/15 down  down     0%    0%      0          0
GigabitEthernet0/0/16 down  down     0%    0%      0          0
GigabitEthernet0/0/17 down  down     0%    0%      0          0
GigabitEthernet0/0/18 down  down     0%    0%      0          0
GigabitEthernet0/0/19 down  down     0%    0%      0          0
GigabitEthernet0/0/20 down  down     0%    0%      0          0
GigabitEthernet0/0/21 down  down     0%    0%      0          0
GigabitEthernet0/0/22 down  down     0%    0%      0          0
GigabitEthernet0/0/23 down  down     0%    0%      0          0
GigabitEthernet0/0/24 down  down     0%    0%      0          0
---- More ----

```

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The total number of vlans is : 3

U: Up; D: Down; TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;

VID Type Ports

1 common UT:GE0/0/5 (D) GE0/0/6 (D) GE0/0/7 (D) GE0/0/8 (D)
 GE0/0/9 (D) GE0/0/10 (D) GE0/0/11 (D) GE0/0/12 (D)
 GE0/0/13 (D) GE0/0/14 (D) GE0/0/15 (D) GE0/0/16 (D)
 GE0/0/17 (D) GE0/0/18 (D) GE0/0/19 (D) GE0/0/20 (D)
 GE0/0/21 (D) GE0/0/22 (D) GE0/0/23 (D) GE0/0/24 (D)

10 common UT:GE0/0/1 (U) GE0/0/2 (U)
20 common UT:GE0/0/3 (U) GE0/0/4 (U)

VID Status Property MAC-LRN Statistics Description

1 enable default enable disable VLAN 0001
10 enable default enable disable VLAN 0010
20 enable default enable disable VLAN 0020

[SW-1-GigabitEthernet0/0/1]

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```
LSW1
LSW1
1    enable default      enable  disable  VLAN 0001
10   enable default      enable  disable  VLAN 0010
20   enable default      enable  disable  VLAN 0020
[SW-1-GigabitEthernet0/0/1]
[SW-1-GigabitEthernet0/0/1]q
[SW-1]q
<SW-1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Dec  9 2025 17:52:59-08:00 SW-1 %%01CFM/4/SAVE(1)[56]:The user chose Y when deciding whether to save the configuration to the device.
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<SW-1>save cun
<SW-1>save cur
<SW-1>save cur
<SW-1>save cur
<SW-1>save cur
<SW-1>save cur
<SW-1>save config.cfg
<SW-1>save config.cfg
Are you sure to save the configuration to flash:/config.cfg?[Y/N]:y
Now saving the current configuration to the slot 0.
Dec  9 2025 17:53:19-08:00 SW-1 %%01CFM/4/SAVE_FILE(1)[57]:When deciding whether to save the configuration to the file flash:/config.cfg, the user chose Y.
Save the configuration successfully.
<SW-1>
<SW-1>
<SW-1>
```

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```
LSW1
LSW1
<SW-1>
<SW-1>start
<SW-1>startup con
<SW-1>startup cu
<SW-1>startup con
<SW-1>startup con
<SW-1>startup ?
    patch                Set patch file
    saved-configuration   Saved-configuration file for system to startup
    system-software       Config system software for system to startup

<SW-1>startup sav
<SW-1>startup saved-configuration config.cfg
<SW-1>startup saved-configuration config.cfg
Info: Succeeded in setting the configuration for booting system.
<SW-1>ls
    ^
Error: Unrecognized command found at '^' position.
<SW-1>dir
Directory of flash:/

   Idx  Attr      Size(Byte)  Date          Time          FileName
   ---  ---
   0    drw-          -   Aug 06 2015  21:26:42   src
   1    drw-          -   Dec 09 2025  17:36:53   compatible
   2    -rw-         503   Dec 09 2025  17:53:00   vrpcfg.zip
   3    -rw-        1,552   Dec 09 2025  17:53:19   config.cfg

32,004 KB total (31,964 KB free)

<SW-1>
```

Name	Date modified	Type	Size
 BE2122DE-1B45-40bb-98C1-ADEF914B323F	12/9/2025 5:54 PM	File folder	
 LAB-01	12/9/2025 5:54 PM	TOPO File	8 KB